

CSci 5707

Principles of Database Systems

Sample Exam 1 Questions

Q1. The owner of several apartment buildings is interested in a database to manage his business.

- Buildings have one or more apartments. Every building has an address.
- Apartments have apt. numbers, and are characterized by their size: 1BR, 2BR, etc.
- Tenants lease apartments. Each lease has a lease date and a period of lease. Tenant information is his/her name and phone number.
- Each building has a manager. The manager has a name, telephone no. and salary.
- Each building has some parking spaces. Some tenants rent the parking spaces.

Design an E-R diagram for the above. State any additional assumptions.

Q2. Convert the above E-R diagram (Q.1) to relational model and write the SQL commands to create the tables for both the entities as well as relationships. State what constraints could not be captured by the tables.

Q3. The following questions are based on the database schema described below. The keys are underlined.

Student(Name, StudentNumber, Class, Major)

Course(CourseName, CourseNumber, CreditHours, Department)

Prerequisite(CourseNumber, PrerequisiteNumber)

Section(SectionIdentifier, CourseNumber, Semester, Year, Instructor)

Grade_Report(StudentNumber, SectionIdentifier, Grade)

(a) Write the following query in SQL:

Retrieve the student number, name and major of all students who do not have a grade of D or F in any of their courses, and sort them by increasing order of student number.

(b) Translate the query of part (a) into a query tree.

(c) Pick a join from the query tree and discuss whether it is better to use Nested-Loops or Sort-Merge join algorithm to evaluate it. Give reasons for your choice.

(d) If you were allowed to create one index, which one would it be to improve the performance of the join in part (c). Explain.

(e) Describe the advantages and disadvantages of having an index on Student Number towards processing this query.

Q4. The following questions are based on a Sporting Goods database described below:

customer (id: *int*, name: *string*, city: *string*, country: *string*, rating: *string*, sales_rep_id: *int*)

dept(id: *int*, name: *string*, region_id: *string*)

sales_rep(id: *int*, last_name: *string*, first_name: *string*, dept_id: *int*, salary: *int*)

Student Name:	Student ID:
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order(id: *int*, customer_id: *int*, date_ordered: *date*, total: *int*)

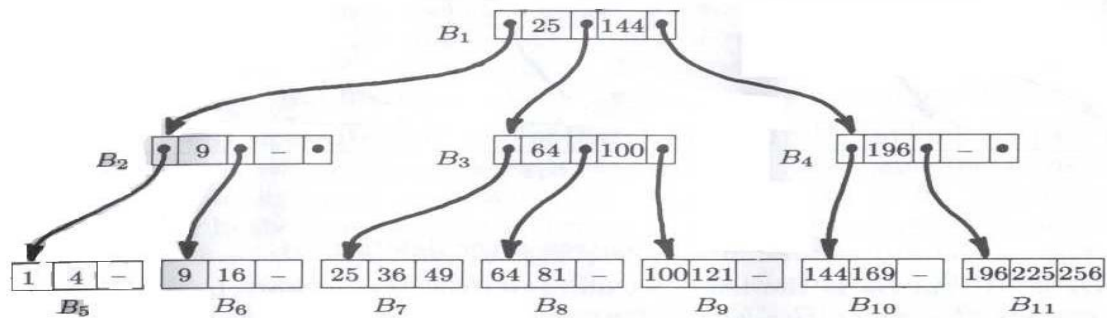
Write SQL queries for each of the following sub-questions.

- (a) Display the name, city, country and rating of all customers whose number of orders exceeds the “average” number of orders for a customer.
- (b) Display the name of all the departments that have at least one employee.
- (c) Display the first name and last name of all sales representatives who do not have customers.
- (d) Find the countries in which there are no sales representatives. If required, make any assumptions and state them.

Q5. For each of the questions below, please keep your answers brief and clear.

- a. Explain the exact meaning of the following ER model notation: ———
- b. What is a “weak entity”? What is a “partial key”?
- c. Define the terms “primary key constraint and foreign key constraint”.
- d. In “cascading delete”, deletion of a referenced tuple causes deletion of all referencing tuples. Why does it not happen the other way around?
- e. Relational calculus is said to be a *declarative* language, in contrast to algebra, which is a *procedural* language. Explain the distinction.
- f. What is a “view”? How do views support logical data independence?

Q6. Consider the B+ tree shown in the figure. Answer the following questions.



- Describe in detail the path taken (i.e. the sequence of comparisons made and links traversed) when searching for a key with value 105.
- Describe in detail the steps in retrieving all records in the range 30 through 150.
- If a B+ tree index matches the selection condition, how does clustering affect the cost?
- You are required to perform a join between relations Reserves and Sailors (on common attribute Sid). Sailors is sorted on Sid, while Reserves has an unclustered index on Sid. Which join algorithm will you select to perform this join (using speed as your criterion). Explain.